TP Ansible

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The objective of this lab is to introduce Ansible and automate the deployment of an architecture consisting of multiple Apache2 servers behind a HAProxy load balancer using Ansible.

Connection

You must be connected either to WifiInp if using your personal computer, or to one of the lab machines at N7.

The lab will be conducted on two virtual machines (VMs). Each student will have access to two VMs, which will form their cluster.

Your supervisor will assign two numbers to you (xx and yy), which you will use to connect to your virtual machines. Throughout the lab, these virtual machines will be referred to as your servers. The password is **toto** for both.

```
ssh -p 130xx ubuntu@147.127.121.11

ssh -p 130yy ubuntu@147.127.121.11
```

Ansible Installation

You will need to modify the hosts file (`/etc/hosts`) and designate one server as the master and the other as the slave. First, retrieve the IP addresses of both servers using the command `ip a` to obtain the IP addresses. Then, update the `/etc/hosts` file accordingly.

This is an example of /etc/hosts content.

```
127.0.0.1 localhost
MasterIP master
SlaveIP slave
....
```

Installing the necessary packages for Ansible.

```
sudo bash
apt-get update
apt-get install sshpass
apt-get install python3
apt-get install python3-pip
apt-get install python3-venv
exit
pip install virtualenv
```

Create a Python virtual environment.

```
source ansible/bin/activate
# Install ansible
pip install ansible
```

Create a file with the list of machines, `nodes.ini`. Here is an example of the content.

```
master
slave
```

ssh key exchange.

```
ssh-keygen -t rsa
ssh-copy-id -i /home/ubuntu/.ssh/id_rsa.pub ubuntu@slave
ssh-copy-id -i /home/ubuntu/.ssh/id_rsa.pub ubuntu@master
```

On each node do this to suppress password

```
sudo visudo -f /etc/sudoers.d/ubuntu
```

Add this content

```
ubuntu ALL=(ALL) NOPASSWD: ALL
```

Check your ansible installation from the master

```
ansible -i nodes.ini -m ping all  --user ubuntu
```

Ansible palybook

First playbook: Installing Python on the slave node.

Create a directory named `playbooks, then create your first playbook,` installPython.yml`.

Here is the content of the file (read and try to understand. If there is any questions, do not hesitate to ask).

```
- name: My first playbook to install python
  hosts: slave # here give your instance name
  user: ubuntu
  become: yes
  tasks:
    - name: Install Python3
      package:
        name: python3
        state: present
```

Run the playbook.

```
ansible-playbook playbooks/installPython.yml  -i nodes.ini -K
```

Install ansible module for docker

```
ansible-galaxy collection install community.docker
```

To install Docker on the slave node using Ansible we will create a playbook named `installDocker.yml`.

Create a file `docker.sh` in the `/home/ubuntu/` directory. The content or this file is:

```
#!/bin/bash
sudo apt-get update -y

sudo apt-get install ca-certificates curl -y

sudo install -m 0755 -d /etc/apt/keyrings

sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o
/etc/apt/keyrings/docker.asc

sudo chmod a+r /etc/apt/keyrings/docker.asc

echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt-get update -y

 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-
compose-plugin -y
```

Here is the content of `installDocker.yml`

```
- name: Playbook to install docker
  hosts: slave
  user: ubuntu
  become: yes
  tasks:
    - name: Update apt repo and cache
      apt: update_cache=yes force_apt_get=yes cache_valid_time=3600
    - name: Copy Docker installation script update script
      ansible.builtin.copy:
        src: /home/ubuntu/docker.sh
        dest: ~/docker.sh
        mode: 0770
    - name: Run Installation of Docker
      command: ~/docker.sh
```

Run the `intallDocker.yml` playbook and check the output.

Let us start a container on the slave using ansible.

Create a new playbook name `startContainer.yml` with the following content and run it.

```
- name: Playbook to install docker
  hosts: slave
  user: ubuntu
  become: yes
  tasks:
  - name: pull an image
    community.docker.docker_image:
      name: ubuntu
      source: pull
  - name: Start a docker container
    community.docker.docker_container:
      name: testdocker
      state: started
      image: ubuntu
      command: /bin/sleep infinity
```

Deploying apache from docker labwork

Now, move back on the docker labwork.

We will build the docker images `apache:v1` form the docker labwork. We create a playbook to create our container from `apache:v1`.

```
- name: Init playbook for apache:v1 image creation
  hosts: slave
  user: ubuntu
  become: yes
  become_method: sudo
  tasks:
    - name: Run container with index.php app
      community.docker.docker_container:
        name: apache
        image: teabetab/apache:v1
        state: healhtly
```

Run the playbook and check the output and access your `index.php` application as we did in the docker labwork.